

Multilinear Interpolation

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1 Questions for Thiele

- how/what kind of multilinear interpolation is used in the multiple ergodic averages
- how important are the bad tuples (quasi banach estimates) in practice? do we need them formalized?
- how important is it to have the optimal bounds on the weak parameters of the multilinear interpolation? does a weaker bound suffices if it means an easier proof?

2 Introduction

These notes offer an overview of some different approaches to multilinear interpolation. They are framed in the setting of [1, Chapter 2], which provides explanation for

the notation and definitions used here and in [2].

We introduce the problem by considering two cases.

2.1 Interpolating between two points

This is the case considered in [2]. It also seems to be used as a last step in the convex hull case, to be covered on section 2.2. Let

$$\bar{A}_1 = (A_0^{(1)}, A_1^{(1)}), \dots, \bar{A}_n = (A_0^{(n)}, A_1^{(n)})$$

and $\bar{B} = (B_0, B_1)$ be Banach couples, and let

$$T: \Sigma(\bar{A}_1) \times \dots \times \Sigma(\bar{A}_n) \rightarrow \Sigma(\bar{B}),$$

be a multilinear function that restricts to two multilinear functions

$$T: A_i^{(1)} \times \dots \times A_i^{(n)} \rightarrow B_i$$

for $i = 0, 1$, such that there are constants K_0, K_1 for which

$$\|T(f_1, \dots, f_n)\| \leq K_i \|f_1\| \cdots \|f_n\|$$

for $f_k \in A_i^{(k)}$ for $1 \leq k \leq n$, and each $i = 0, 1$.

Given $\theta \in (0, 1)$ and numbers $q_1, \dots, q_n, q \in [1, \infty]$ satisfying some specified conditions, we would be interested in showing that T also makes sense as a bounded multilinear operator between the intermediate subspaces given by the real interpolation method:

$$T: (\bar{A}_1)_{\theta, q_1} \times \dots \times (\bar{A}_n)_{\theta, q_n} \rightarrow (\bar{B})_{\theta, q},$$

with some constant K depending on K_1 and K_2 .

Example 2.1. If our Banach couple $\bar{A} = (A^{(0)}, A^{(1)})$ is given by Lebesgue spaces $A^{(0)} = L^{p_0, q_0}$, $A^{(1)} = L^{p_1, q_0}$ and $\underline{p}_0 < p_1$, then the real interpolation space corresponds to the Lebesgue space $L^{p, q} = (\bar{A})_{\theta, q}$, where $p^{-1} = (1 - \theta)p_0^{-1} + \theta p_1^{-1}$.

2.2 Interpolating on the interior of a convex set

We assume that the Marcinkiewicz real interpolation theorem for linear operators is available, and we see how it can generalize to the multilinear case.

[3]

3 Applications

3.1 Usage in the commuting ergodic averages paper

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References

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